

LAB ASSIGNMENT -1

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Q1)

// Menu driven program to display the following array operations

//MENU

//CREATE , DISPLAY , INSERT , DELETE , LINEAR SEARCH , EXIT

#include <iostream>

using namespace std;

int main() {

int arr[100], n = 0; // array and size

int choice;

do {

cout << "\n---- MENU ----\n";

cout << "1. CREATE\n";

cout << "2. DISPLAY\n";

cout << "3. INSERT\n";

cout << "4. DELETE\n";

cout << "5. LINEAR SEARCH\n";

cout << "6. EXIT\n";

cout << "Enter your choice: ";

cin >> choice;

if (choice == 1) {

cout << "Enter number of elements: ";

cin >> n;

cout << "Enter " << n << " elements:\n";

```

    for (int i = 0; i < n; i++) {
        cin >> arr[i];
    }
    cout << "Array created.\n";
}

else if (choice == 2) {
    if (n == 0) cout << "Array is empty.\n";
    else {
        cout << "Array elements: ";
        for (int i = 0; i < n; i++) {
            cout << arr[i] << " ";
        }
        cout << endl;
    }
}

else if (choice == 3) {
    int pos, val;

    cout << "Enter position to insert (1 to " << n+1 << "): ";
    cin >> pos;

    cout << "Enter value: ";
    cin >> val;

    if (pos < 1 || pos > n+1) {
        cout << "Invalid position!\n";
    } else {
        for (int i = n; i >= pos; i--) {

```

```

        arr[i] = arr[i-1];
    }
    arr[pos-1] = val;
    n++;
    cout << "Value inserted.\n";
}
}
else if (choice == 4) {
    int pos;
    cout << "Enter position to delete (1 to " << n << "): ";
    cin >> pos;
    if (pos < 1 || pos > n) {
        cout << "Invalid position!\n";
    } else {
        for (int i = pos-1; i < n-1; i++) {
            arr[i] = arr[i+1];
        }
        n--;
        cout << "Value deleted.\n";
    }
}
else if (choice == 5) {
    int key, found = 0;
    cout << "Enter value to search: ";
    cin >> key;

```

```
for (int i = 0; i < n; i++) {  
    if (arr[i] == key) {  
        cout << "Value found at position " << i+1 << ".\n";  
        found = 1;  
        break;  
    }  
}  
if (!found) cout << "Value not found.\n";  
}  
else if (choice == 6) {  
    cout << "Exiting program...\n";  
}  
else {  
    cout << "Invalid choice!\n";  
}  
} while (choice != 6);  
return 0;  
}
```

OUTPUT

```

---- MENU ----
1. CREATE
2. DISPLAY
3. INSERT
4. DELETE
5. LINEAR SEARCH
6. EXIT
Enter your choice: 1
Enter number of elements: 6
Enter 6 elements:
12
24
36
48
52
64
Array created.

---- MENU ----
1. CREATE
2. DISPLAY
3. INSERT
4. DELETE
5. LINEAR SEARCH
6. EXIT
Enter your choice: 2
Array elements: 12 24 36 48 52 64

---- MENU ----
1. CREATE
2. DISPLAY
3. INSERT
4. DELETE
5. LINEAR SEARCH
6. EXIT
Enter your choice: 3
Enter position to insert (1 to 7): 1
Enter value: 2
Value inserted.

---- MENU ----
1. CREATE
2. DISPLAY
3. INSERT
4. DELETE
5. LINEAR SEARCH
6. EXIT
Enter your choice: 4
Enter position to delete (1 to 7): 5
Value deleted.

---- MENU ----
1. CREATE
2. DISPLAY
3. INSERT
4. DELETE
5. LINEAR SEARCH
6. EXIT
Enter your choice: 5
Enter value to search: 6
Value not found.

---- MENU ----
1. CREATE
2. DISPLAY
3. INSERT
4. DELETE
5. LINEAR SEARCH
6. EXIT
Enter your choice: 6
Exiting program...

-----
Process exited after 69.16 seconds with return value 0
Press any key to continue . . . |

```

Q2)

//Program to delete vowels from the string

#include <iostream>

```
using namespace std;

bool isVowel(char ch) {
    return (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u' ||
            ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U');
}

int main() {
    string str, result;

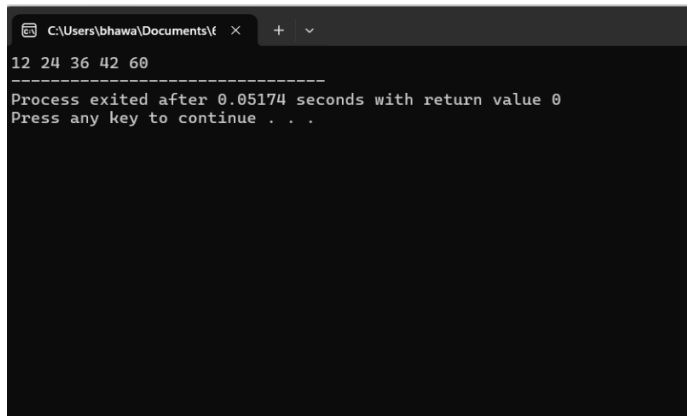
    cout << "Enter a string: ";
    getline(cin, str);

    for (size_t i = 0; i < str.length(); i++) {
        if (!isVowel(str[i])) {
            result += str[i];
        }
    }

    cout << "String after deleting vowels: " << result << endl;

    return 0;
}
```

OUTPUT



```
C:\Users\bhawa\Documents\{ } x + v
12 24 36 42 60
-----
Process exited after 0.05174 seconds with return value 0
Press any key to continue . . .
```

Q3)

//Predict the output

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    int i;
```

```
    int arr[5]={1};
```

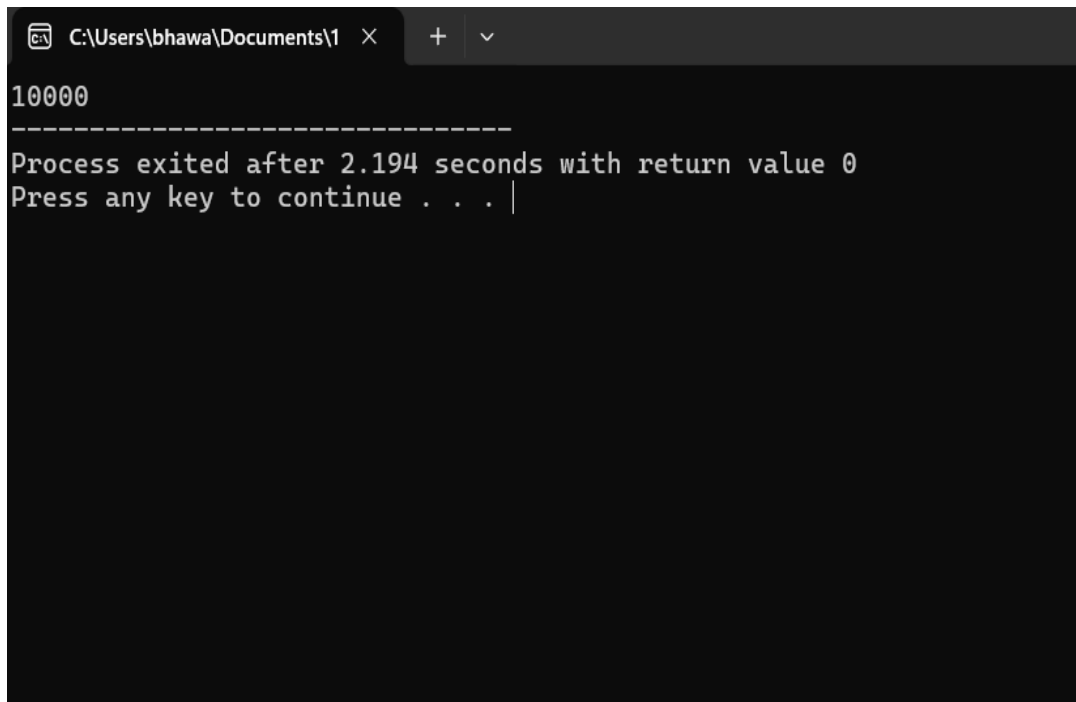
```
    for (i = 0; i<5; i++)
```

```
        printf( "%d", arr[i]);
```

```
    return 0;
```

```
}
```

OUTPUT



```
C:\Users\bhawa\Documents\1  X  +  v
10000
-----
Process exited after 2.194 seconds with return value 0
Press any key to continue . . . |
```

Q4) a

//Program to Reverse elements of an array

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int n;
```

```
    cout << "Enter the size of the array: ";
```

```
    cin >> n;
```

```
    int arr[n];
```

```
    cout << "Enter " << n << " elements: ";
```

```
    for (int i = 0; i < n; i++) {
```

```
        cin >> arr[i];
```

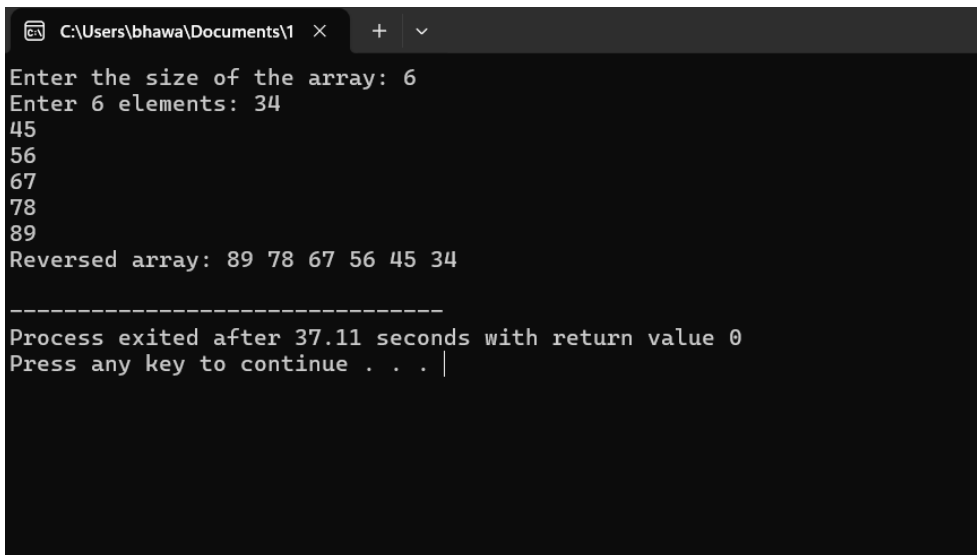
```
    }
```

```

// Reverse the array
for (int i = 0; i < n / 2; i++) {
    int temp = arr[i];
    arr[i] = arr[n - i - 1];
    arr[n - i - 1] = temp;
}
cout << "Reversed array: ";
for (int i = 0; i < n; i++) {
    cout << arr[i] << " ";
}
cout << endl;
return 0;
}

```

OUTPUT



The screenshot shows a Windows command prompt window with the following text:

```

C:\Users\bhawa\Documents\1 x + v
Enter the size of the array: 6
Enter 6 elements: 34
45
56
67
78
89
Reversed array: 89 78 67 56 45 34
-----
Process exited after 37.11 seconds with return value 0
Press any key to continue . . . |

```

b)

//Program to find the matrix multiplication

```
#include <iostream>
```

```
using namespace std;
```

```

#define R1 2 // number of rows in Matrix-1

#define C1 2 // number of columns in Matrix-1

#define R2 2 // number of rows in Matrix-2

#define C2 3 // number of columns in Matrix-2

void mulMat(int mat1[100][C1], int mat2[100][C2])
{
    int rslt[R1][C2];

    cout << "Multiplication of given two matrices is:\n";
    for (int i = 0; i < R1; i++) {
        for (int j = 0; j < C2; j++) {
            rslt[i][j] = 0;

            for (int k = 0; k < R2; k++) {
                rslt[i][j] += mat1[i][k] * mat2[k][j];
            }

            cout << rslt[i][j] << "\t";
        }

        cout << endl;
    }
}

// Driver code

int main()
{
    // R1 = 4, C1 = 4 and R2 = 4, C2 = 4 (Update these
    // values in MACROs)

    int mat1[R1][C1] = { { 4, 4 }, { 6, 6 } };

    int mat2[R2][C2] = { { 4, 4, 4 }, { 6, 6, 6 } };

    if (C1 != R2) {

        cout << "The number of columns in Matrix-1 must "

```

```

        "be equal to the number of rows in "
        "Matrix-2"
    << endl;

    cout << "Please update MACROs according to your "
        "array dimension in #define section"
    << endl;

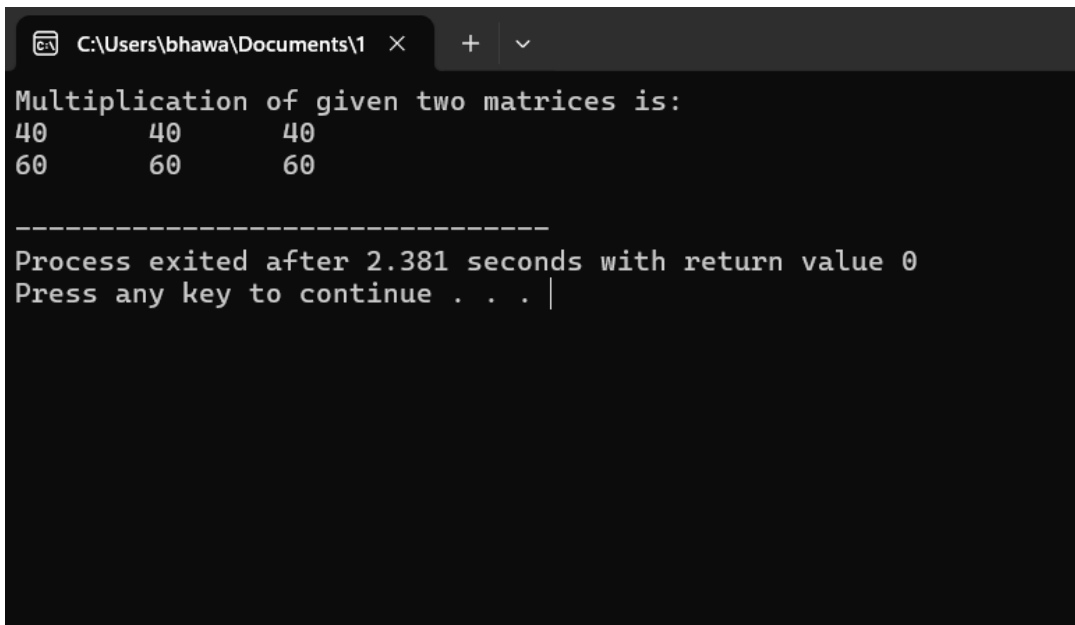
    exit(EXIT_FAILURE);
}

// Function call
mulMat(mat1, mat2);

return 0;
}

```

OUTPUT



```

C:\Users\bhawa\Documents\1 >
Multiplication of given two matrices is:
40      40      40
60      60      60

-----
Process exited after 2.381 seconds with return value 0
Press any key to continue . . . |

```

C)

```

//Program to find transpose of a matrix

#include <iostream>

```

```

using namespace std;

// This function stores transpose
// of A[][] in B[][]

void transpose(int A[][N],
               int B[][N])
{
    int i, j;

    for (i = 0; i < N; i++)
        for (j = 0; j < N; j++)
            B[i][j] = A[j][i];
}

int main()
{
    int A[N][N] = {{1, 1, 1, 1},
                   {2, 2, 2, 2},
                   {3, 3, 3, 3},
                   {4, 4, 4, 4}};

    int B[N][N], i, j;

    transpose(A, B);

    cout << "Result matrix is \n";

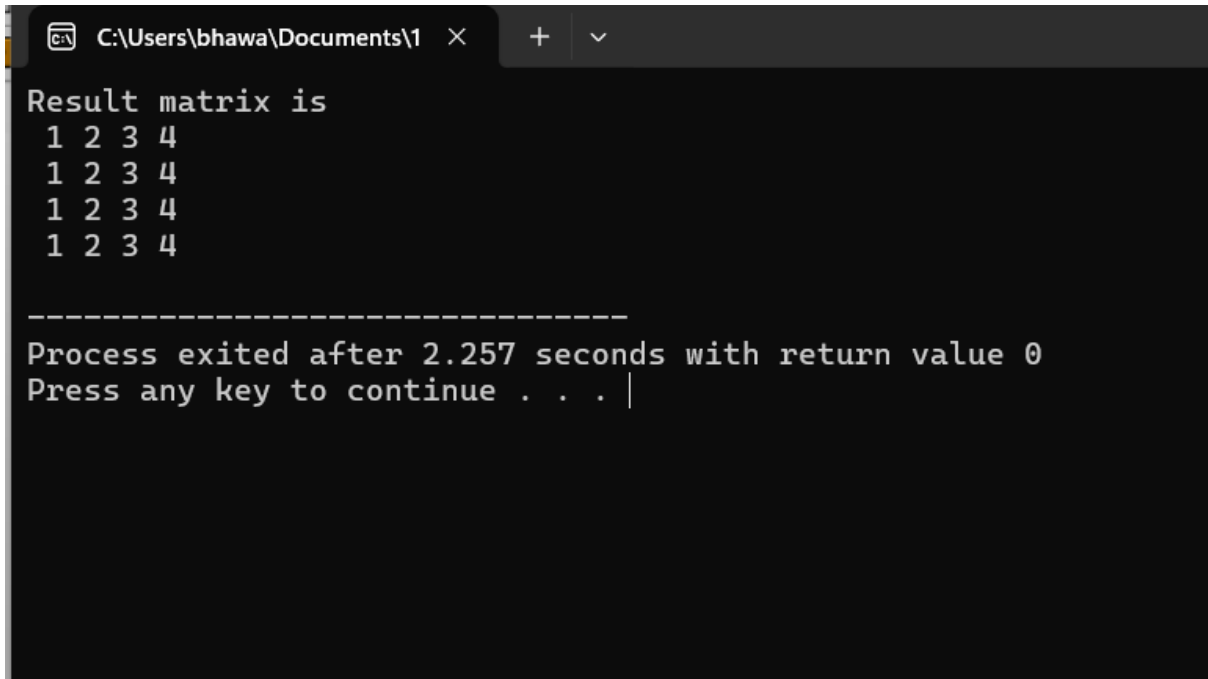
    for (i = 0; i < N; i++)
    {
        for (j = 0; j < N; j++)
            cout << " " << B[i][j];

        cout << "\n";
    }

    return 0;
}

```

OUTPUT



```
C:\Users\bhawa\Documents\1 × + v
Result matrix is
1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4

-----
Process exited after 2.257 seconds with return value 0
Press any key to continue . . . |
```

Q5)

//Program to find sum of every row and column in two dimensional array

```
#include<iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    // Define the dimensions of the 2D array
```

```
    const int ROWS = 3;
```

```
    const int COLS = 4;
```

```
// Declaring and initializing the 2D array
```

```
    int matrix[ROWS][COLS] = {
```

```
        {8, 9, 10, 11},
```

```
        {12, 13, 14, 15},
```

```
        {16, 17, 18, 19}
```

```
    };
```

```

// Calculate sum of row

std::cout << "Row Sums:" << std::endl;

for (int i = 0; i < ROWS; ++i) {

    int rowSum = 0;

    for (int j = 0; j < COLS; ++j) {

        rowSum += matrix[i][j];

    }

    std::cout << "Sum of Row " << i + 1 << ": " << rowSum << std::endl;

}

// Calculate sum of columns

std::cout << "\nColumn Sums:" << std::endl;

for (int j = 0; j < COLS; ++j) {

    int colSum = 0;

    for (int i = 0; i < ROWS; ++i) {

        colSum += matrix[i][j]; //To find the missing number in sorted array
    }

}

#include <iostream>

using namespace std;

int findMissing(int arr[], int n) {

    int diff = arr[0]; // Expected difference (since array should be continuous)

    for (int i = 0; i < n; i++) {

        if (arr[i] - i != diff) {

            return i + diff;

        }

    }

    return -1; // No missing element

}

int main() {

```

```

int arr[] = {1, 2, 3, 4, 6, 7, 8};

int n = sizeof(arr) / sizeof(arr[0]);

int missing = findMissing(arr, n);

if (missing != -1)

    cout << "Missing element is: " << missing << endl;

else

    cout << "No element is missing." << endl;

return 0;
}

}

std::cout << "Sum of Column " << j + 1 << ": " << colSum << std::endl;

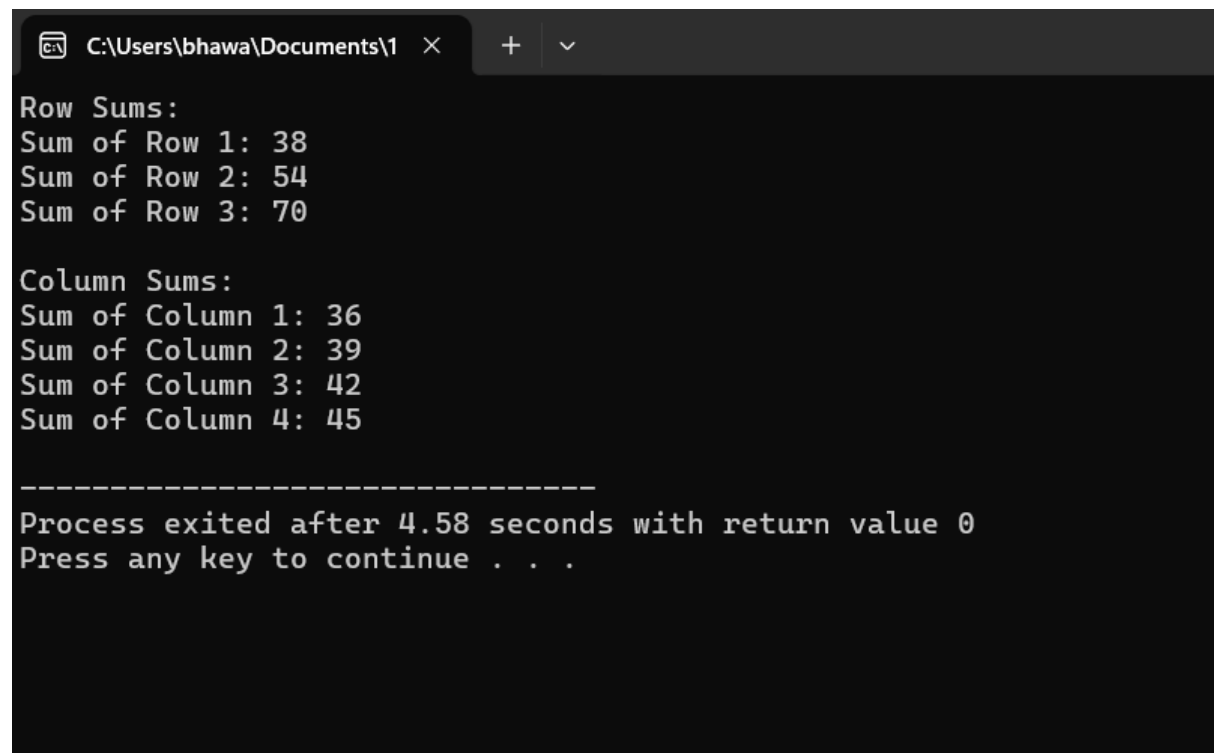
}

return 0;

}

```

OUTPUT



```

C:\Users\bhawa\Documents\1 x + v
Row Sums:
Sum of Row 1: 38
Sum of Row 2: 54
Sum of Row 3: 70

Column Sums:
Sum of Column 1: 36
Sum of Column 2: 39
Sum of Column 3: 42
Sum of Column 4: 45

-----
Process exited after 4.58 seconds with return value 0
Press any key to continue . . .

```

